

SCD FINAL EXAM: MASTER REVISION GUIDE

1. SOFTWARE DESIGN PROCESS (CHAPTER 2)

Definition: Transforming customer requirements (SRS) into a form implementable by code.

The 3 Phases of Software Design:

Interface Design

Architectural Design

Detailed Design

7 Software Design Concepts:

Abstraction, Modularity, Architecture, Refinement, Pattern,

Information Hiding, Refactoring

2. SOLID DESIGN PRINCIPLES

SRP, OCP, LSP, ISP, DIP

3. DESIGN PATTERNS

Creational: Singleton, Factory

Behavioral: Observer, Strategy

4. CODE QUALITY & METRICS

LOC, Cyclomatic Complexity, Code Coverage, Coupling, Cohesion, Technical Debt

5. REFACTORING & READABILITY

14 Refactoring Techniques

6. FUNDAMENTALS OF TESTING

Unit, Integration, System, Acceptance

TDD: Red → Green → Refactor

7. DEBUGGING

Identify, Analyze, Fix, Test, Document

8. EXCEPTION HANDLING & LOGGING

try, catch, finally, throw

Logging Levels: DEBUG, INFO, WARN, ERROR, FATAL

9. SOFTWARE DEPLOYMENT

Blue-Green, Canary, Rolling

CI/CD Pipeline

10. SOFTWARE MAINTENANCE

Corrective, Adaptive, Perfective, Preventive

11. CODE SALVAGING & CONFIGURATION MANAGEMENT

Version Control, Change Control, Build Management, Release Management